

VENKATA DOGIPARTHY

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SUMMARY

Computer Science student with hands-on experience developing AI-driven applications and data pipelines. Skilled in Python, C++, prompt orchestration with LangChain, and vector storage for LLM-based retrieval. Built end-to-end offline AI chatbots and deployed prototypes on Windows and AWS environments. Eager to support exploratory AI/ML research and multimodal system development in an IRAD setting.

TECHNICAL SKILLS

- **Programming & Development:** Python, C#, C++, JavaScript
- **Web Development:** ASP.NET Core MVC, React.js, HTML5, CSS3, Bootstrap
- **Databases & Storage:** SQL Server, MongoDB, ChromaDB, Firebase
- **Cloud & Deployment:** AWS, IIS Webserver, Vercel
- **Backend & Architecture:** Entity Framework Core, LINQ, RESTful APIs, ASP.NET Identity, Role-Based Access Control, Data Validation
- **Communication & Integration:** SMTP Email Integration, Third-Party APIs (Google API)
- **Tools & Platforms:** Git, Github, Visual Studio, Visual Studio Code
- **AI & Machine Learning:** Large Language Models (LLMs), LangChain, Retrieval-Augmented Generation (RAG), Vector Databases, Data Cleaning
- **Mobile Development:** Android App Development, React Native

WORK EXPERIENCE

Astute Solutions

Aug 2025 - Dec 2025

AI Engineer Intern

- Developed an end-to-end chatbot leveraging large language models (**LLaMA**) using Python-based Streamlit and Ollama, implementing multi-turn conversational memory for fully offline inference.
- Cleaned and preprocessed internal documents before integrating them into **ChromaDB** for vector storage and semantic search, enabling retrieval-augmented responses with optimized prompt design.
- Built a secure, local-first design without APIs for ensuring data privacy, low latency, and functional performance, and used LangChain for prompt orchestration in artificial intelligence workflows.
- Tested, debugged, and deployed the application independently for Windows OS and AWS Cloud to control the model setup and environment.

University of South Alabama

Oct 2024 - Aug 2025

Auxiliary Police

- Supported campus security, emergency response, and event operations.
- Assisted in law enforcement responsibilities and student safety procedures.
- Participated in community outreach and safety programs, building positive relationships between police and students.

DigiFrills IT Solutions

Nov 2023 - Mar 2024

Software Developer Intern

- Collaborated with a four-person development team to build and deploy a feature-rich CRM application using React, improving customer data management.
- Developed dynamic forms for key CRM functionalities, ensuring smooth data entry and integration with backend APIs.
- Led QA testing efforts, performing manual and automated tests to identify and resolve issues, ensuring forms met quality and security standards.

Web Developer Intern

- Created interactive web pages such as a calculator, reminders application, and tribute page using HTML, CSS, and JavaScript.
- Implemented simple logic in JavaScript for interactions and dynamic changes.
- Focused on frontend UI design, layout structuring and usability improvements.

ACADEMIC PROJECTS

Alumni Management System

- Developed a full-scale web application for managing alumni records, academic history, employment, and institutional communication using **ASP.NET Core MVC** and **SQL Server**.
- Designed a structured onboarding workflow where alumni accounts are created only after verification against a pre-imported institutional **registry**, preventing duplicate accounts.
- Implemented **bulk data import functionality** to process large alumni datasets, **intelligently handling duplicate records** by adding new degree entries instead of overwriting existing data.
- Built **dynamic reports** module(Alumni Count, Average GPA, Mailing Labels) with filtering, grouping, and Excel export to support data-driven decision making.
- Integrated **SMTP**-based email functionality to handle password reset workflows and securely deliver temporary credentials to registered alumni and announcements module enabling administrators to send system-wide notifications and updates to alumni directly through the platform and through email.
- Enforced robust **validation** across the system including GPA ranges, year constraints, and strict data formatting to ensure high data integrity.
- Implemented **role-based access** control to separate admin and staff functionalities, improving system security and usability.
- Deployed the application on an **IIS web server** with SQL Server backend, enabling secure internal access within the university environment.

Construction Management System | [Demo](#)

- Developed a completely functional web application based on **CRUD**, as per the requirements of construction site management.
- Designed a **role-based UI** to provide personalized access to users of different types.
- Added functionality to the **dashboard** to provide progress bars that enable easy viewing of project/task progress with a responsive UI using Bootstrap and CSS with light/dark mode.
- Improved collaboration among Site Managers, Property Owners, and Workers through the provision of personalized dashboards for work monitoring and property management.

Senior Capstone Management System | [Demo](#)

- Created a fully functional web application using CRUD for professors to manage projects for students.
- Added a role-based user interface and allocated corresponding access privileges for administrators and viewers.
- Automated distribution of project survey links using a Node.js script and JavaScript, reducing manual outreach by 90%
- Built a Python-based data pipeline with pandas for CSV import and data cleaning, eliminating manual entry and improving data consistency for student records
- Implemented the system on the internal server of the university using IIS for safe and seamless logins for authorized personnel.

Lawn Mowers Database - Backend System

- Designed an optimized database for a lawn mower business.
- Implemented many-to-many relationships with efficient query handling.
- Ensured smooth integration with frontend applications.
- Delivered consistent results even under complex query loads.

Product Shelf-Life Reminder

- Developed a mobile application for tracking product expiration dates with automated device notifications to reduce waste and improve inventory awareness.
- Implemented **Firebase Authentication** for secure and scalable user data management.
- Designed a responsive mobile UI with category-based organization, filtering, and search functionality.
- Integrated external APIs (Google Search API) to enhance product lookup and data retrieval.
- Configured push notifications using Firebase Cloud Messaging to alert users about upcoming expirations.

CERTIFICATIONS

- Intel FICE Software Certification
- Microsoft Certified: Azure Fundamentals (AZ-900)
- Microsoft Certified: Power Platform Fundamentals (PL-900)
- AWS Cloud Computing

EDUCATION

University of South Alabama <i>Master of Science, Information Systems</i> GPA: 3.60	Aug 2024 – May 2026 <i>Mobile, Alabama</i>
Parul University <i>Bachelor of Technology, Computer Science</i> GPA: 8.66	Nov 2020 - May 2024 <i>Vadodara, Gujarat</i>